The call control protocol in a separated call and bearer environment

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The separation of call and bearer to support future teleservices and services in broadband networks is outlined in this paper. Various mechanisms to undertake the separation are overviewed and the method chosen by the International Telecommunications Union (ITU) and the European Telecommunications Standards Institute (ETSI) is described in detail. A full protocol model for broadband access signalling is derived and explained from the basic steps leading through to the establishment of an association between end-points supporting call control information flows. The call control protocol, currently being developed to establish, maintain and clear a call with more than one connection, is outlined and illustrated by information flows.

1. Introduction

Broadband networks are the future platforms for telecommunications companies and these networks will only be economically viable if they support a diverse range of teleservices [1], providing scope for the developers of applications. The provision of generic communications systems has long been the aim of public network operators (PNOs), and diverse communications options complicate the problem of controlling a network independently of the application. Any complex problem is simplified by breaking it down into constituent parts and, in the case of a diverse broadband network, the International Telecommunications Union — Telecommunications Standardisation Sector (ITU-T) has decomposed the problem into the control of the connection by a bearer control and the co-ordination of a series of related connections into the rather more abstract mechanism known as call control. The ability of a separated call control to communicate the requirements for a call has been recognised as sufficiently complicated to warrant a new protocol. This protocol has been developed to provide a generic communication structure that is independent of the network and protocol transport system. ITU-T have separated the features that they require in the signalling systems into capability sets. This paper concentrates on the features required of the call control protocol to enable broadband signalling capability set 2 (CS-2) to be implemented. Further information can be found elsewhere on the capability set approach by ITU-T [2], and the services and signalling requirements [1] .

2. The separation of call and bearer

The narrowband telephony network can be regarded, at least as far as the signalling is concerned, as supporting two very similar teleservices. The voice teleservice provides a bi-directional symmetrical connection between two terminals guaranteed to transfer commercial speech band (300 Hz to 3.1 kHz) analogue frequency signals, while the data teleservice guarantees to make digital, synchronous bi-directional, symmetrical connections at 64 kbit/s. Neither service, however, is application specific — a group 3 fax machine is an example of an application that takes advantage of the facilities offered by the voice teleservice, or analogue connection, to provide a service that is entirely different to human voice communication. The commonality between the two underlying teleservices lies in the fact that both services provide a single bi-directional symmetrical connection between two end-points through one or more networks. This description of the communications could be referred to as a ‘call’ and is set up implicitly during the request for connectivity.

Broadband networks, and asynchronous transfer mode (ATM) in particular, offer many more diverse types of connections. The support for constant bit rate, variable bit rate and the ability to separately specify the peak cell rate, sustainable cell rate and many other additional characteristics of an ATM connection [3] enable diverse applications to use the same communications network. If those applications are combined together, such as a video-telephony call with additional data exchange (e.g. a shared whiteboard for 'scribbling space'), the connections need to be co-ordinated into a single relationship — the 'call'. This ensures that additional connections are not offered as a new call but set up within the restrictions of the existing call.

2.1 Functional model

A full separation of protocols was first identified in ITU-T signalling requirements in an unpublished draft recommendation Q.73, which described the ITU-T's functional model of processing entities involved in a call and a simplified model is reproduced in Fig 1. Separating call control and bearer control protocols provides support for the future convergence of the fixed and mobile networks, the Universal Mobile Telecommunications System [4], as well as negotiating network and bearer resources to prevent needless reservations in calls failing due to terminal incompatibilities. This is increasingly likely in broadband networks, due to the diverse terminals, services and information encodings that may be supported. Additional advantages include reducing the redundant processing of call control information at intermediate bearer control nodes (relay bearer control nodes (RBC)). By including information directing the protocol messages towards a call control entity, the RBCs are spared from decoding and 'understanding' the whole of the message and its parameters.

The functional model uses the concept of a terminal, or other user equipment as taking on the role of either originating or destination signalling end-point, and the signalling entities within the terminal are referred to as control agents. Splitting the functionality into call and bearer control results in an originating call control agent (OCCA) and originating bearer control agent (OBCA) in the call originator's terminal, and similarly a terminating call control agent and terminating bearer control agent in the destination user equipment. The functional entities in the network nodes show call control entities only at the edges of the network. The ITU-T model is more general and does allow call control entities at other points, such as at gateway locations or other network boundaries. In the centre of Fig 1 the relay bearer control (RBC) node provides the procedures to support switching connections but has no concept of the call and so cannot support service functionality. This means that, for example, it can route connections only to the next bearer control node.

2.2 Controlling connections

Bearer control is defined as providing the procedures to enable a connection to be set up, maintained, modified and released. The bearer control is responsible for the overall control of individual connections, the control of the network resources and the collection of traffic count information. Bearer control is also sometimes known as link control or connection control and is based upon the signalling protocols mainly developed in the ITU-T and published as Recommendations Q.2931 [5], Q.2971 [6], and B-ISUP [7] (see also Jones [8]). The overall control of a connection is performed by the protocol undertaking the set-up and release procedures associated with a connection and the maintenance of the state of the connection. Connection control is performed across a network and between networks, selecting the connection identifier and protecting networks (e.g. by maintaining timers). Controlling network resources is performed at each node by the reservation and release of network resources, node selection and negotiation, resource interworking (e.g. with existing networks), and the through connection and disconnection of bearers to commence and cease communications. These functions are common to both broadband and narrowband networks, but for broadband networks there are two new resources to control — the parameters concerned with the bandwidth and the overall quality of service of a connection, and the functions specific to the handling of connection branches for multipoint connections. The ability to support connections of differing characteristics introduces two concepts new to broadband networks — the

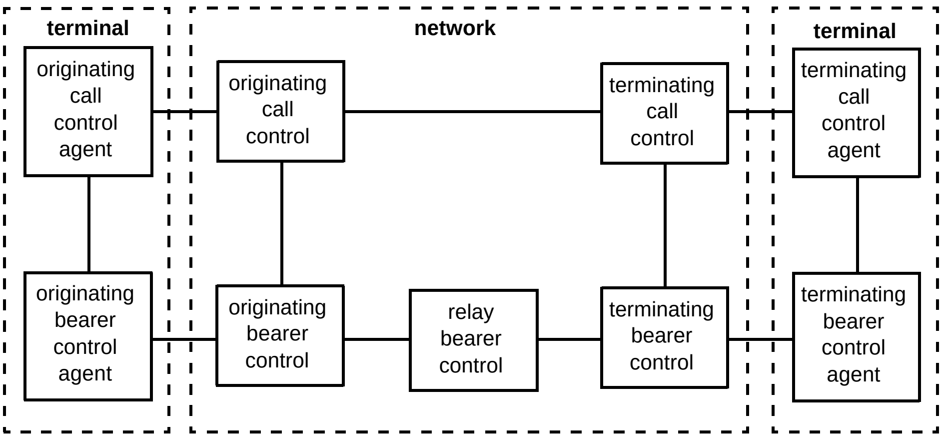

Fig 1 Simplified functional model of processing entities.

procedures concerned with both the negotiation and the modification of a connection's characteristics [9, 10].

2.3 Controlling calls

In separating the call from its connections, the ITU-T has replaced the old definition used to describe the monolithic control of the connection and its embedded call with a new definition of call control:

> "Call control is a signalling association between one or more terminals (user applications) and the network to control the set-up, release, modification and maintenance of sets of communication connections to provide related information. Call control is used to maintain the association between parties, and a 'call' may embody any number of underlying connections, including zero, at any instance in time."

Call control can be realised in a number of ways:

- separating the call information into parameters carried by a single call/bearer protocol,
- separating the state machines for a call control and a bearer control while signalling information in a single call/bearer protocol,
- performing a full separation of information and state machines by providing separate signalling protocols.

The full separation of call and bearer protocols enables a number of functions to be realised that applications can use to make communications more efficient. The call can be set up to enable the characteristics of the communications session (the 'service') to be established first and the connection characteristics to be determined before the connection is established. This is referred to as call negotiation and connection prenegotiation.

The full separation of the call control and the bearer control protocols facilitates a simpler upgrading strategy, and allows the separation of the call control software from the telecommunications switches. This provides a step towards making the service provisioning independent from the switching technology and the suppliers by enabling the separation of the call control protocol states from the information it is trying to convey.

3. Separation of protocol and information

Traditionally, signalling protocols have reflected the states of the call and its embedded connection, reusing names in the protocol states and messages. To take one example, communication with another network end-point (a 'party') is requested by sending a 'SETUP' message; the destination party may inform the originator that the user has been contacted by sending an 'ALERTING' message, and the confirmation that the communication channel is available is indicated by a 'CONNECT' message. The states in which the call is placed reflect the status of the destination party, although to a lesser extent. The called party's protocol states associated with the three messages above are 'call present', 'call received' and 'active.'

Broadband calls in CS-2 [2] have the ability to support multiple end-points (multiparty calls) and the states and messages become less clear. In CS-1 protocols it is an error to put multiple destination party numbers in a 'SETUP' message. A multiparty call must allow this situation, and the consequential state explosion of attempting to reflect the states of all the parties becomes overly complex. A new approach was needed for any type of call supporting more than one connection or more than one party, or both. The approach adopted was to define a protocol that carries the information, concerning the changes of states of parties and connections, as parameters to a small set of protocol messages. Storing and manipulating the state changes that the protocol carries is more efficiently performed by applying object-oriented techniques, leaving the protocol states concerned only with the progression of any related protocol actions.

The separation of the protocol from the information it carries can be represented by the general schematic for an object-model-driven protocol shown in Fig 2. The user application manipulates the object model via an application programming interface (API) and the object model signals any changes through the protocol (in this case the call control protocol) to the network. The network has a similar arrangement, with the changes being recorded within an object model which then updates the exchange application process (in this case the call control service user application — CCSUA). More details on the requirements for using an object-oriented model as the basis of a call control protocol to support multiparty scenarios is given in McDonald [11] and some service scenarios for multiparty multiconnection calls are given in Morley and Knight [1].

4. Modelling a call with objects

The object model is concerned with the abstraction of a call into its fundamental parts and the status of thos e parts. A complex engineering problem, as this is, can always be simplified by breaking the problem up into constituent parts. A call has a number of constituent parts, each requiring its own particular data. This type of situation

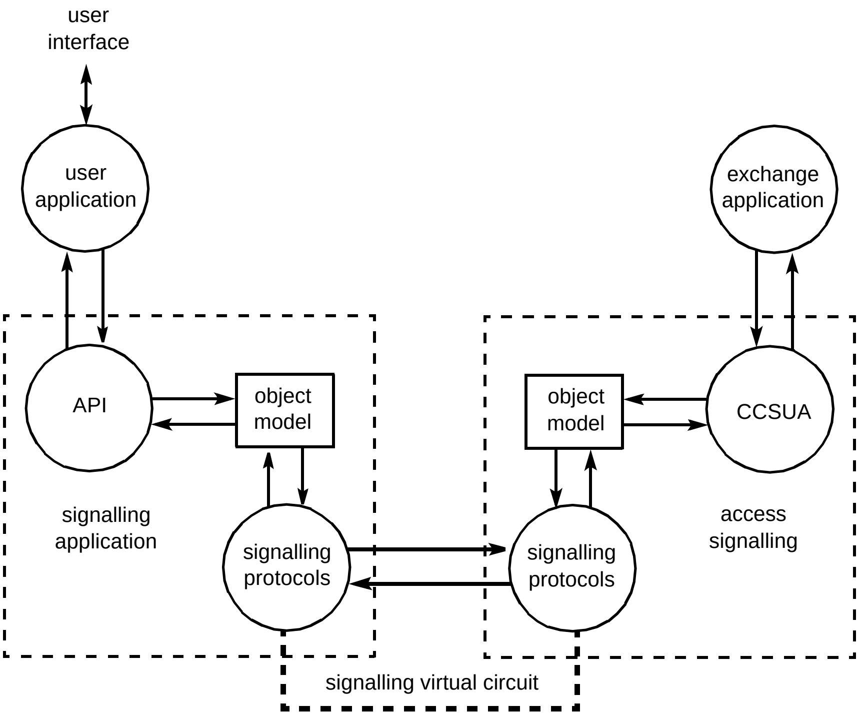

Fig 2 Signalling state changes at the user-network interface through an object model.

readily lends itself to a description based upon object modelling, and the objects that are required to describe the call as an information model are shown in Fig 3.

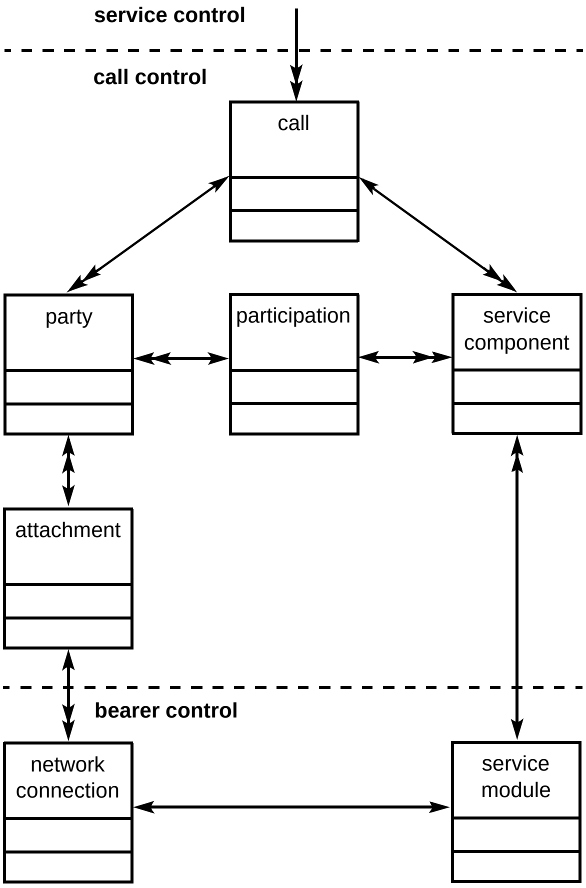

Fig 3 Basic information object model.

4.1 *Call objects*

The model divides the signalling objects into three types — call control objects, service control objects and connection control objects. Service control objects, which are left undefined in CS-2, are concerned with abstraction into attributes of a service. An example of the use of separation of service from the call control is the DAVIC specified 'session control', described in Miles et al [12]. Call control associated objects represent the aspect of a call that involves end-to-end associations between parties and the network that are independent of the other characteristics of the service. Call control objects refer to particular connections but do not store the characteristics of the connections, for example, connection object identities need to be 'known' but the bandwidth and the other quality of service characteristics of the connection are not defined in call control.

4.2 *Service component and service module objects*

Bearer control associated objects are directly associated with the communications capabilities of the connections and are modelled as two object types. The network connection object provides the containment of the connection's attributes, such as peak cell rate, sustainable cell rate, etc. The service module represents the hardware that a terminal may use to interpret the bit stream provided by the ATM adaptation layer. To illustrate this with an example, a bit stream may consist of video pictures (in some format) and sound (in some format). These two formats are combined together and compressed to produce a Moving

Picture Experts Group (MPEG) stream which may be carried in a single ATM virtual channel as shown in Fig 4. The type of multiplexing and the compression algorithm applied need to be signalled to ensure that both terminals can communicate, but the information does not need to be interpreted or checked by the network. This demonstrates the difference between the service module and the connection object — the attributes of the connection object must be understood by terminals and the network, while the service module is information that needs to be shared by the terminals during the connection set-up. The bearer control protocol is heavily based upon the CS-1 Q.2931 protocol, which does not signal information using an object model, but uses more traditional information elements. In the example shown in Fig 4, sound and pictures are identified, but the terminals will also need to agree to use the same coding algorithm — such as μ-law, A-law, etc, for sound. This does not have the same sort of effect on the connection characteristics as the service module used. Terminals may wish to determine that the basic service is compatible from one end to the other, before setting up connections, by signalling the details using the service component object.

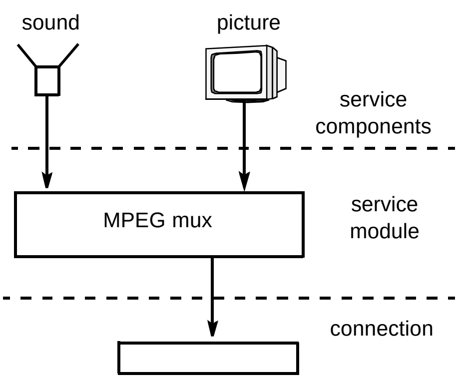

Fig 4 Service components.

4.3 Party object

Call control is concerned with providing generic communications between conceptual end-points (parties) and a network to implement a service. The call control level must therefore identify who is participating in the call, through the party object. A party may have been proposed to be part of the call or may have confirmed their involvement in the call. The status of the party object may therefore be ‘virtual’ until that party has confirmed participation.

4.4 Call object

The call object provides the binding together of the parties and service components to the connections by containing data to identify the other objects in the information model. It must also identify which party this call control entity is co-ordinating, i.e. the ‘local party.’ The other parties in the call are regarded as remote parties. This means that a single call control entity is not expected to co-ordinate more than one local party. The inclusion of multiple remote parties — held in the call object as a list of references to party objects — is included for the future, when multiple parties will be able to communicate over a variety of connections within a single call.

4.5 Participation and attachment objects

In a complex call it is necessary to identify what information each of the parties is sending and receiving. This is identified in an object model by defining a link, or association, between two objects. It becomes particularly difficult to define a multiple association of the type generally referred to as ‘many-to-many’. An example of this is the association between parties and service components. Figure 4 illustrates the situation where a single party must be associated with a number of service components (two in this case — but it could be more). Each service component can be expected to have at least two parties associated with it — a sender and a receiver — but in the future it will be possible to have more than two. The commonest approach in object modelling for resolving such a ‘many-to-many’ link is to introduce another, simpler, object that references pairs of objects to associate the two together. The participation object resolves the many-to-many association between the party and the service component object, while the attachment object performs a similar association between a party and a connection.

4.6 Connection object

The objects associated with bearer control are concerned with the control of the connection, and included in the model for reference. Bearer control is not implemented as an object model, and the separation of connection and service module is therefore theoretical. In practice, the call control can refer to either a connection or a service module through a single object reference, the connection object ID, which is defined by call control (by allocating a unique value) and carried by the bearer control protocol.

5. Protocol architecture

Signalling using an object model requires a number of functions to interact in a co-ordinated manner. Bearer control interactions cannot take place until the call and, as has already been mentioned, the connection prenegotiation procedures in a call, have been established. These are just some of the operations that need to be supported in the signalling protocols; others include supplementary services and a means of transporting the call control operations. The mechanisms that bind these functions together to enable them to co-operate in various procedures is represented in a protocol architecture model. This type of signalling communication readily lends itself to an application layer structure [13] which enables functions to be used as necessary, and ignored if not needed. For example, prenegotiation is only an option to be used as necessary.

Figure 5 shows how the signalling protocols interact in a local broadband exchange (an ATM switch). Any modern telephone exchange is dependent on the software that implements the various functions that the exchange undertakes. Many of these functions are either specific to the particular switch manufacturer, or are specific to the particular network operator. These functions, such as call records for charging mechanisms, network management, testing, etc, are exchange application processes, which are not a subject for standardisation and not covered in this paper. Access signalling and, more generally, the user/network interface handler, whose applications are present on the left side of Fig 5, have been split into two parts.

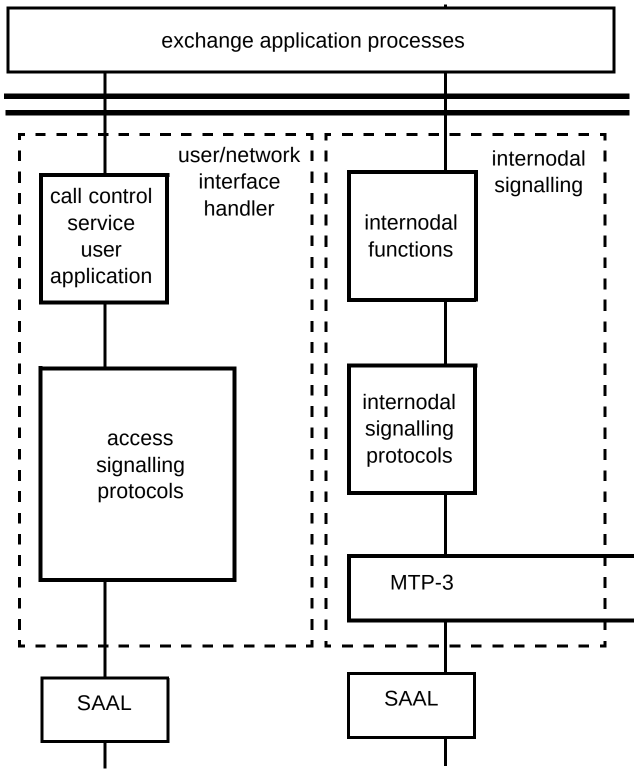

Fig 5 General architecture for broadband signalling.

The call control service user application provides that part of the access signalling that is concerned with the current state of the parties and the connections in the call. The CCSUA is not concerned with the details of the signalling communication, but only with the current status of the terminal (connected, answering, etc) and its current attributes (for example, the address or number), the service and any supplementary services, and similar details for the connections.

The access signalling protocols support the CCSUA by providing the means for the states and attributes (data) held in the CCSUA to change. The access signalling protocols free the CCSUA from the more mundane actions concerning the procedures for establishing, maintaining and releasing a communications association.

This general architecture enables the information concerning the call to be separated from the protocol, allowing for appropriate methods of specification to be applied. Traditionally the CCSUA has not been a subject for standardisation. The transport of the information that enables the object model to be shared among CCSUA entities involved in the call must, however, be standardised to enable communication between equipment realised by different suppliers. The information ultimately takes the form of parameters to the access signalling protocols, providing the object references and their attributes.

5.1 Access signalling protocols

The access signalling protocols provide the means to exchange the information between terminals and networks. Since broadband calls are not always simple point-to-point, mono-connection calls, the signalling will also need to support the call consisting of a single point-to-multipoint connection. This type of call utilises the ATM capability of multicast (often incorrectly termed a broadcast call), in which cells are copied to more than one destination at a time by the switch. To co-ordinate actions between the parties in such a call, a function for party control is specified. There are also functions to support restarting the protocol under error conditions, such as link failures detected by the SAAL or corruption of data within an exchange application process. The full protocol architecture for the user/network interface handler is shown in Fig 6.

The structure of the architecture for the access signalling elements is based upon the B-ISUP approach [14], which pre-dated the access signalling discussions in ITU-T and took into account the architecture framework using open services interconnection (OSI) concepts [13]. The actions that the various elements undertake are linked together to form a set of co-operating procedures by a control function (CF). This type of architecture, consisting of a control function and any number of application service elements (ASEs), follows the concepts outlined in ITU-T [15] and the International Standards Organisation (ISO) [16] of an application layer structure (ALS). This structure specifies that an application service object (ASO) consists of a CF + ASEs + ASOs, which is a recursive definition. This allows for extensions, if necessary, in CS-3 [2] for multiparty support. For certain CS-3 calls, especially complex multiparty, multiconnection calls, it is not clear whether a single control function is adequate. However, it is anticipated that the most drastic change will be to add a further ASO, although the current work on the protocol

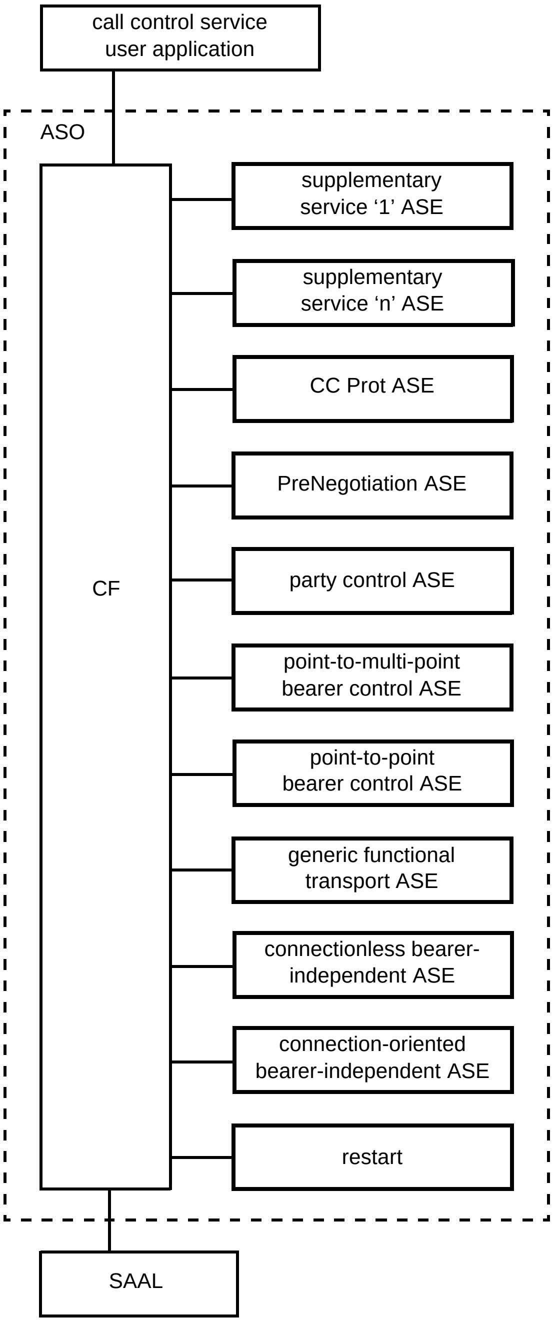

Fig 6 Access signalling application service elements.

requirements for CS-3 have not identified the need for another ASO to be added.

Interfaces between the individual ASEs and CF are primitive interfaces. The interface between the CF and signalling ATM adaptation layer (SAAL) [17] is the SAAL service interface [18], described in Law and Knight [2]. All functions also have an interface to a 'management application', although this has not been defined as a formal primitive interface.

Supplementary service ASEs can be expected to be added as necessary. The implementation of supplementary services is generally dependent upon the network operator and the capabilities in the terminal. For example, a service such as 'transfer on no reply' may be considered to be appropriate for the emulation of a narrowband type private network, but is less likely to be found in public networks.

ITU-T have identified that prenegotiaton could be applied both to call and connection attributes. While this is true in principle, at present no call prenegotiation has been identified, since the negotiation of the services required within a call may be adequately negotiated during call establishment. For multiconnection calls, a prenegotiation ASE has been defined to enable this operation to be undertaken either at call establishment, or subsequently — prior to the establishment of a connection. Although it is possible that prenegotiation might be considered as a sub-operation of call control, ITU-T were of the opinion that the operation is sufficiently complex to warrant a separate protocol, and a 'stand-alone ASE' approach has been adopted.

Party control was first defined as a mechanism within point-to-multipoint connection control, but that was an interim solution to enable simple multicast to be defined very early on. ITU-T accepted restructuring for a harmonised architecture even though party control could also be considered as a sub-operation of call control. As with prenegotiation, a stand-alone party control ASE is a better approach, providing a simpler and more generic solution.

The bearer control consists of a number of components — point-to-point, point-to-multipoint, modification and negotiation of traffic characteristics (Q.2961 [3]), etc — and, while it is possible to make each of these separate ASEs for definition within the protocol architecture, a generic mechanism is preferable, enabling a single BC ASE to be both comprehensive and simple. Only in the case of the combined call and bearer control (Q.2931 [5]) and point-to-multipoint (Q.2971 [6]) protocols are stand-alone ASEs appropriate.

5.2 *Internodal signalling*

The signalling protocol between the nodes in a network (nominally switches, or exchanges) is known as internodal signalling. Since service functions are invoked at the edges of the network (access and gateway points), there is no requirement for call control functionality at these points. ITU-T's B-ISUP protocol [8] controls the connections within a call, but does not need to co-ordinate the connections. This principle can also be applied to the ATM Forum protocol, PNNI [19]. For more details of both of these protocols, see Jones [8] and Scott and Jones [20].

6. Call control protocol actions

Two of the major requirements for the call control protocol are the co-ordination of connections (multiconnection calls), and the co-ordination of parties (multiparty calls). Since call control is 'a signalling association between co-ordinating entities,' a single call control protocol has been derived that is generic enough to satisfy both these cases. The simpler case is the co-ordination of multiple connections within a call. This has the advantage, from the network provider's point of view, of reducing the number of times that a routeing address needs to be derived from the end-point address supplied by the originating terminal. To co-ordinate multiple connections, the following actions need to take place within the network:

- all connections must be routed to pass through every co-ordinating (call control) entity,
- call control associations must be completed before bearers are established,
- a call may exist without any bearers (facilitating complex connection rearrangements).

The second requirement is for call control to co-ordinate the actions of multiple parties. This is achieved by using the concept of atomic actions, coupled with a protocol that allows the principle of 'undoing' a request if one party refuses a request. An atomic action is a means of breaking a large number of interactions into a set of smaller, inter-related actions. Multiparty principles and atomic actions are described in McDonald [11].

6.1 Call establishment

The call control service user application provides the means for a serving node co-ordinating a call to hold an object model representation of the call. In practice, the node can only 'see' those parties and connections with which the local party is associated. This is known as the 'local view'. The global, or complete, view of a call may not exist at all — although it helps us to understand how the object modelling works — but it is not necessarily required by any node in the call control.

The call control protocol is responsible for communicating the details of the basic information model between co-ordinating entities (call control network nodes). In a simple two-party call, this can take place by a simple request and response interchange to communicate the local view of the call at establishment, as shown in Fig 7.

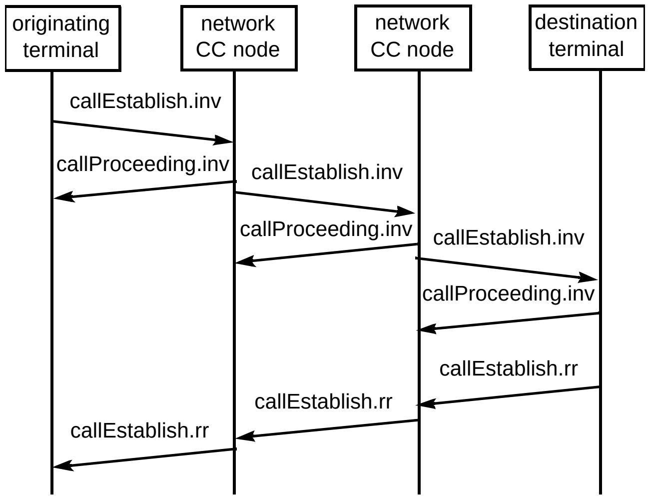

Fig 7 Call establishment in call control protocol.

The flows do not show messages between call control co-ordinating entities, since messages must also pass through the intervening relay bearer control (RBC) nodes as shown in the functional model of Fig 1. Instead, the flows represent the call control protocol operations, which are parameters to the underlying messages. Information on the call control operations must be contained (enveloped) within an underlying protocol transport mechanism to enable the call control protocol flows to pass between nodes in different parts of the network. This is covered later in this paper, as are the '.inv' and '.rr' extensions in the flows.

It has already been shown that the CC entities are not necessarily adjacent, and so the mechanism for making the signalling association is slightly more complicated than with the current DSS2 or B-ISUP. DSS2 uses the connection reference, unfortunately named the call reference information element in Q.2931, to provide the association for bearer control by indicating which end of the link allocated the unique number used to direct messages to the ASO.

In the CC protocol, as there may be more than one call control node beyond the end of a bearer link (see Fig 8), two call establishment requests received over a common signalling link may both have the same referencing number. The mechanism adopted is for the originating CC to allocate half the reference number (a unique number so far as that CC is concerned) and the destination CC to allocate the other half. The call proceeding flow is used by the destination CC to indicate to the originator its half of the reference. This reference number becomes unique between a pair of CC entities and is therefore named the call segment identifier.

The call proceeding flow is also used to confirm the routeing address of the CC entity, enabling a call to start setting up connections prior to the addressed terminal being contacted.

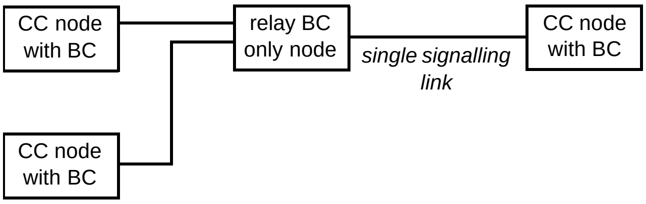

Fig 8 Call control co-ordinating nodes.

6.2 Bearer establishment

Figure 7 does not show the establishment of any connections. The exact point at which connections are established must be service dependent — and can even be application dependent. It is the implementor's choice to decide whether a particular service will be best implemented by setting up the connections as soon as possible, or whether some connection prenegotiation is more appropriate. In order to emulate the narrowband ISDN (and POTS) service, ITU-T at first placed a requirement that the first connection can be set up simultaneously with the call. This has caused some heated discussion, but it has been shown that this is not a real requirement. The real requirement is for the connection to be available at the time that the user is offered the call, otherwise 'clipping' may occur. Clipping is a phenomenon due to our habit of picking up a ringing telephone's handset and immediately speaking to the caller. If the connection is not immediately available, then some loss of speech can result, dependent on the connection set-up delay. A simple solution is to wait until the connection is available before the presence of the call is indicated to the user. This may be added into the diagram as shown in Fig 9.

The flows show both call control protocol flows and bearer control protocol messages. Bearer control messages are indicated by dotted lines and the BC initials precede the message name. For simplicity the flows represent a two-party call between terminals serviced by two adjacent broadband exchanges.

The originating terminal chooses to implement the fast set-up of the first connection by performing a BC set-up as soon as the callProceeding.inv is received. The network has to delay the initial address message (IAM) until the callProceeding.inv is received from the destination local exchange. The connection is accepted across the network using the initial address acknowledge (IAA) and the set-up is sent to the terminal. The destination terminal chooses not to offer the call to the user until the first connection is available. The user is made aware of the presence of the call by some terminal-dependent means and the call control co-ordinating nodes are informed of the change of status of the party by means of an update to the information model in the callStatus.inv flow. This unconfirmed flow updates the status attribute of the party object to indicate that it is now in the 'alerting' state. In the case shown, the call is accepted and the connection completed in the last two messages.

Flows to enable further connections to be set up do not require any interactions at the call control level, since the call is now established and the protocol interactions take place only at the bearer control level.

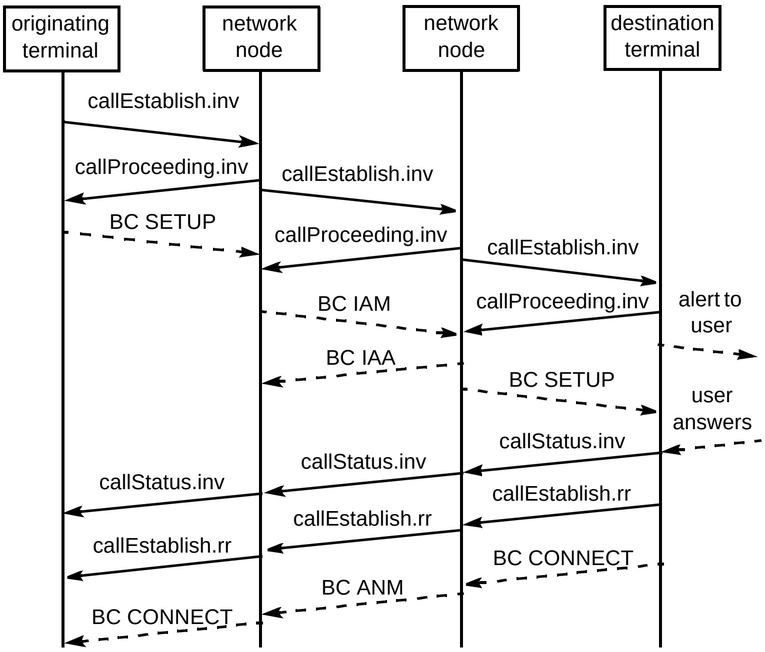

Fig 9 Delay of alerting indication to user.

6.3 Support for multiparty calls

To enable a smooth transition from CS-2 to CS-3, the call control protocol incorporates a single flag that enables it to indicate whether the signalling flow needs to be co-ordinated. These actions support the requirements of a signalling protocol that can 'undo' actions in some parties of a multiparty call when another party is unable to accept a request. The requirements for multiparty calls came out of the RACE MAGIC project, described in McDonald [11]. The CS2 call control protocol may, therefore, be used by a terminal to act as a terminating party in a multiparty call, provided that that party does not communicate directly with more than one of the other parties in the call. Under these circumstances, a co-ordinated approach needs to be taken, similar to the 'BEGIN', 'READY' and 'COMMIT' in RACE MAGIC [21]. While the message names differ, with the new call control protocol using callEstablish.inv, callEstablish.rr and callComplete.inv for BEGIN, READY and COMMIT respectively, the actions and underlying procedures are the same. Figure 10 shows the flows for two segments of a multiparty call, allowing a participating CS-2 destination party. The multiparty co-ordination is not shown in this flow.

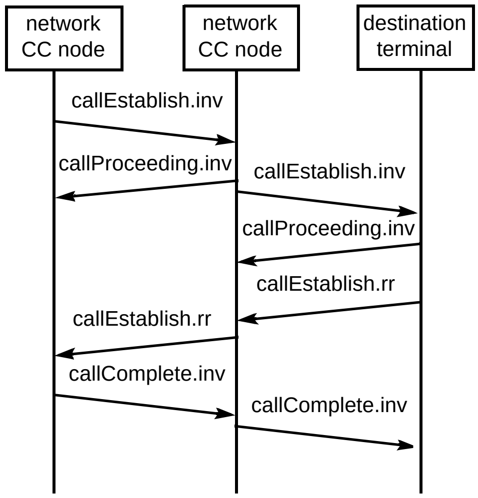

Fig 10 Co-ordinated call control establishment.

6.4 Releasing a call

The release mechanism is shown in Fig 11 using the same mixture of call control and bearer control flows as in Fig 9. The example chosen in Fig 11 is the release of a call with two connections being initiated by call control.

The requirements for multiconnection calls stated that a call may exist without any connections, but that a connection may not exist if it is not contained within a call. The release procedures are always local and cannot fail (which means that any signalling entity may not refuse a release request) and so the call does not clear until an entity's co-ordinated connections have been released. It is not necessary, however, to receive a confirmation of the call's release from a succeeding entity before confirming release to a preceding entity (in either call or bearer

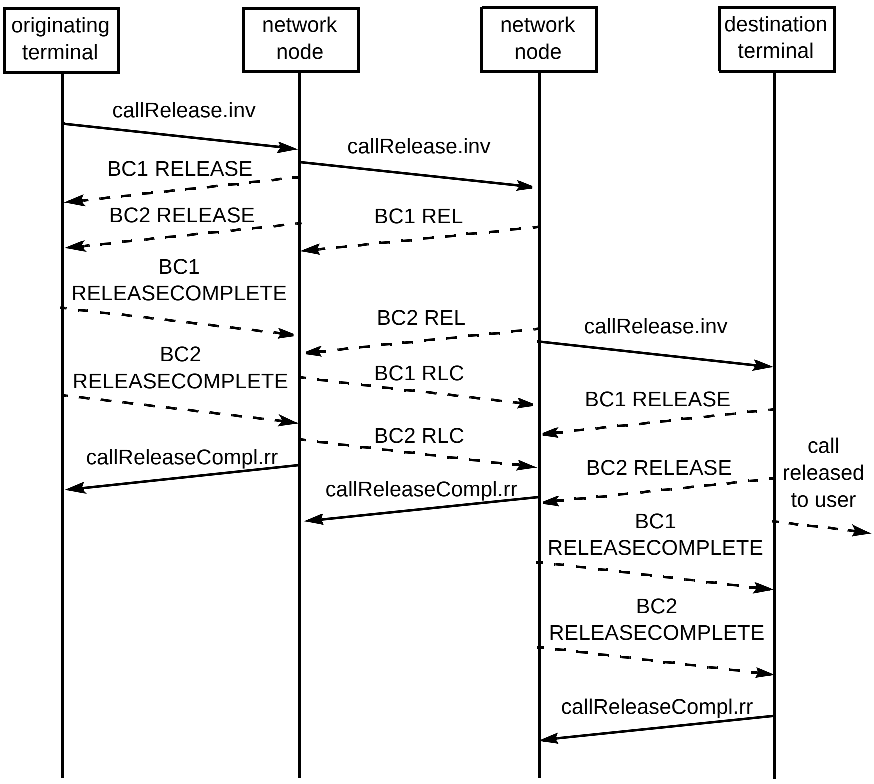

Fig 11 Release of a call with two connections.

control). These requirements are a logical extension of the narrowband and CS-1 signalling protocols. It is envisaged that in CS-3 the only change will be for a party who requests a call release in a multiparty call may merely result in releasing themselves and not the entire call.

7. Protocol transport mechanism

The protocol transport mechanism for call control operations expected to be used by ITU-T is the remote operations service element (ROSE) [22] and is enabled by the generic functional transport mechanism [23]. This mechanism takes advantage of the bearer control protocol (Q.2931 based) to send a connection-oriented bearer-independent (COBI) set-up message. The message contains all the required fields to be acted upon by the receiving bearer control signalling entity except that it does not request a connection. Contained within the fields is a facility information element, which is passed on unchanged until a call control co-ordinating entity is reached. The control function of the ASO (see Fig 6) splits the contents of the message to the appropriate ASE, as shown in Fig 12. The advantage of the GFT protocol is that it can be used as part of a private network system, for example, using the private network signalling protocol, B-QSIG, described in Allard [24]. The ROSE functions can also make use of the transaction capabilities application part (TCAP) [25] facilities of Signalling System No 7 (SS7) [26] to support the transfer of information between non-adjacent signalling entities within a public network.

The ROSE functions support a generic message interchange of an invoke procedure and a confirming return-result, as indicated in Figs 7 and 9 by the '.inv' and '.rr' extensions. A return-result is not always required and so ROSE provides the capability for the invoking entity to indicate whether the operation is confirmed or unconfirmed. The call proceeding is an unconfirmed flow, since a confirmation is not required in this case.

The GFT protocol in broadband access signalling, and the ROSE mechanisms were all imported, largely unchanged, from narrowband signalling. The techniques were first developed to enable supplementary services to be transported, and more details of these generic mechanisms can be found in Griffiths [27].

8. Conclusions

The support of complex services, made possible by ATM, will require communications interactions over more than one channel. The support of such requirements has

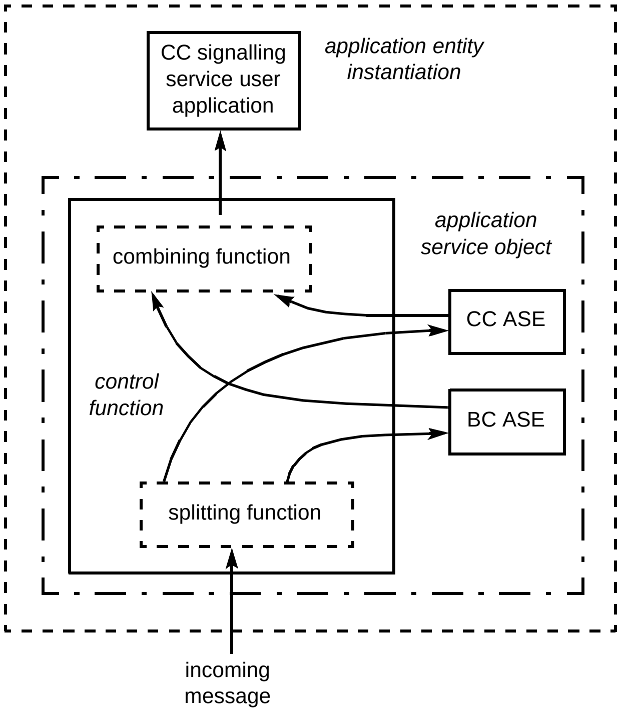

Fig 12 Splitting an incoming message to the ASEs.

been shown to be simplified by separating the functions required for the control of the call from the functions of the control of the connection. By extending this separation into the protocols, it is simpler to signal the state changes of a particular connection within a call. A further separation of the information that the call control protocol signals from the states in which the protocol may be placed will enable a smooth transition in the future for the support of multiparty calls. In adopting such a strategy, the standards bodies have accepted the strategic results of the RACE MAGIC project [21].

Further work has enabled the call control protocol to remain independent of the protocol transport mechanism and allows call control entities to be situated at the point where co-ordination is required, without regard for the underlying signalling protocol stack or the network type.

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